

Ho Chi Minh City Housing Market Analysis

District-Level Price Trends, Inflation Adjustment, and Geographic Drivers | 2017–2022

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DATA300 | May 2026

1. Introduction

This report presents the findings of a data-driven analysis of residential housing prices across nine districts in Ho Chi Minh City (HCM), Vietnam, covering daily observations from January 2017 to mid-2022. The primary aim is to describe and explain district-level price differences using exploratory data analysis (EDA), inflation adjustment, and geographic context. A supplementary forecasting experiment is also included and discussed with its limitations clearly stated.

The nine districts studied are Districts 1, 2, 3, 4, 5, 6, 7, 8, and 9, covering a range of urban typologies from the central business district to peripheral residential areas. All prices are denominated in millions of Vietnamese Dong (VND).

2. Data

Four datasets were used:

- **Primary dataset:** HousePricingHCM.csv —

Daily housing prices for nine HCM districts (1,363 observations per district). This is the core dataset for all analysis.

- **Vietnamese CPI:** consumerPriceIndex.csv —

Annual monthly-average Consumer Price Index values from 1995 to 2024, used to deflate nominal prices into real (inflation-adjusted) terms with 2017 as the base year.

- **World Bank population data:** populationData.csv —

National Vietnam population figures used to compute year-over-year growth rates.

- **District geographic data:** district_context.csv —

District-level area (km²), population, and coordinates, used to compute population density and straight-line distance from each district centroid to District 1 (the CBD).

3. Exploratory Data Analysis

3.1 District Price Levels

Average housing prices vary substantially across districts. District 1, the CBD, commands the highest average at approximately 193 million VND, while District 9 averages just 50 million — a nearly four-

fold difference within the same city. Districts 1, 3, and 5 form a clear premium tier, while Districts 2, 4, 6, 7, 8, and 9 constitute a more affordable band.

District	Avg Price (M VND)	Cumulative Growth (%)	Real Growth (%)
District 1	193	144%	111%
District 2	67	91%	65%
District 3	160	40%	21%
District 4	94	30%	12%
District 5	157	141%	108%
District 6	88	38%	19%
District 7	75	379%	313%
District 8	78	112%	84%
District 9	50	190%	151%

Table 1: Average price and cumulative growth by district. Real growth is CPI-adjusted with 2017 as base year.

3.2 Inflation Adjustment

Adjusting for inflation using the Vietnamese CPI reveals that nominal growth figures materially overstate real appreciation for several districts. Districts 3 and 6 show nominal growth of 38–40% but real growth of only 19–21% — meaning more than half of their apparent price increase was driven by general inflation rather than housing market appreciation. By contrast, District 7 shows 379% nominal growth and 313% real growth, indicating that the overwhelming majority of its gains reflect genuine demand-driven appreciation rather than a monetary effect. This distinction has direct implications for how districts should be evaluated as investment markets.

Key EDA Finding: Inflation erases most gains in Districts 3 and 6

Anyone citing the 38–40% nominal growth figures for Districts 3 and 6 as evidence of strong market appreciation is overstating the case. After CPI adjustment, these districts barely outpaced inflation. District 7 is the only district where the headline growth figure holds up in real terms.

3.3 District Geographic Characteristics

A district context dataset was constructed using area (km²), population, and geographic coordinates to calculate population density and straight-line distance from each district to District 1. These variables show strong relationships with both average price and growth rate:

- Distance to District 1 vs average price: correlation = -0.76 . This is the strongest structural relationship found in the dataset — farther districts are systematically cheaper.
- Population density vs average price: correlation = $+0.51$. Denser districts tend to command higher prices, though this effect is partially confounded with distance to CBD.
- Population density vs cumulative growth rate: correlation = -0.52 . Lower-density districts grew faster over the study period. This relationship is central to understanding District 7.

3.4 District 7: A Case Study in Geographic Drivers

District 7 is the most analytically interesting district in the dataset. Its 313% real growth — the highest of any district — is explained by a specific combination of geographic characteristics:

- Location: 5.3 km from the CBD — close enough to benefit from urban spillover from premium Districts 1, 3, and 5, but affordable enough in 2017 to attract buyers priced out of the center.
- Density: 10,091 people/km² — the lowest population density of any inner district, signaling large amounts of undeveloped land available for construction.
- Area: 35.7 km² — more than four times larger than Districts 3, 4, or 5. Growth from a low starting price across large available land produces dramatic percentage appreciation.

District 7 sits above the price-distance trend line — priced higher than its distance alone would predict — and is a clear outlier on the growth-distance scatter plot. Its position in the K-means cluster analysis (isolated from both the premium and mid-market groups) further confirms its unique profile.

3.5 Price Inequality Across Districts

The absolute price gap between the most and least expensive districts grew from approximately 85 million VND in early 2017 to 147 million VND by mid-2022. However, the coefficient of variation — a measure of relative spread — remained essentially flat throughout the period (approximately 0.43–0.50). This indicates that relative inequality between districts is stable; the premium tier is pulling ahead in absolute VND terms because its prices are higher to begin with, not because the proportional structure of the market is changing.

4. Predictive Modeling (Exploratory)

Important caveat on this section

The following forecasting results are exploratory. The models rely primarily on past prices as predictors and use limited additional variables. Results should not be interpreted as a robust forecasting system and are presented to illustrate both the potential and the clear limitations of price-level prediction with this dataset.

4.1 Price Level Prediction

Three models were trained on a time-based 80/20 train-test split to predict next-day housing prices: Linear Regression, Random Forest, and XGBoost. Features included lag values (previous 1, 2, 3, 7, 14, and 30 days), rolling means and standard deviations, and time variables (year, month, day of year).

Model	MAE (M VND)	RMSE (M VND)	R ²
Linear Regression	1.53	1.90	0.9989
XGBoost	1.58	2.09	0.9986

Model	MAE (M VND)	RMSE (M VND)	R ²
Random Forest	1.79	2.59	0.9979

Table 2: Price-level prediction results on the test set.

While R^2 exceeds 0.99 for all models, this figure is misleading as a measure of model insight. Feature importance analysis reveals that the previous day's price (lag_1) accounts for approximately 52% of XGBoost's total model weight, and lag features collectively account for over 57%. The high accuracy primarily reflects the autocorrelated, smooth nature of daily housing prices — not the discovery of complex predictive structure. In essence, the model is largely learning that tomorrow's price will be close to today's.

Forecast accuracy degrades predictably with horizon: MAE rises from 1.65 million VND at one-day ahead to 2.79 million at thirty days ahead, confirming that short-term price smoothness is being exploited rather than long-run market dynamics being modeled.

4.2 Growth Rate Prediction

An attempt was made to predict 30-day percentage price growth — a harder and more economically meaningful task than predicting the price level. XGBoost was used as the representative model. The results are sobering:

Model / Baseline	MAE (percentage points)
XGBoost	2.17
Zero-growth baseline	2.18
Past-growth baseline	3.31

Table 3: 30-day growth rate prediction results.

XGBoost marginally outperforms a naive zero-growth baseline (2.17 vs 2.18 percentage points MAE), a difference that is not practically meaningful. This result is consistent with the efficient market hypothesis — if short-term price appreciation were reliably predictable from past prices alone, that information would already be priced in. Future work should incorporate supply-side variables (construction permits, available land), interest rates, transaction volume, and infrastructure investment data to improve the explanatory power of growth rate models.

5. Limitations

- The predictive models rely almost exclusively on past prices. External economic drivers — interest rates, housing supply, transaction volume, infrastructure investment — are absent from the forecasting models.
- District-level population data is national in origin (World Bank), meaning population density variation across districts reflects the supplementary geographic dataset rather than time-varying local demand signals.
- K-means clustering is performed on nine data points, which is statistically too few for robust cluster validation. Results are descriptive and illustrative rather than inferential.

- Population density and distance to CBD are correlated with each other ($r = -0.51$), making it difficult to cleanly separate their independent effects on price without a multivariate regression controlling for both simultaneously.
- The dataset covers 2017–2022 and may not reflect conditions after mid-2022, particularly given broader macroeconomic changes in Vietnam’s property market from 2022 onwards.

6. Conclusions

This project’s most substantive contributions are in the EDA rather than the predictive modeling. Three findings stand out as genuinely new and directly supported by evidence:

- District 7’s exceptional growth has a structural geographic explanation — low starting density, large available land, and a location close enough to the CBD to benefit from urban expansion — not just favorable market conditions.
- Inflation adjustment materially changes the investment narrative. Districts 3 and 6 present as moderate performers once CPI is applied, erasing the misleading impression given by nominal growth figures.
- Short-term price appreciation is not reliably predictable from past prices alone. A well-specified model with clean data barely outperforms assuming zero growth, suggesting that supply-side and macroeconomic variables are necessary for meaningful forecasting.

Overall conclusion

Ho Chi Minh City housing prices show strong short-term persistence, making near-term price-level prediction technically accurate but analytically shallow. The more meaningful contribution of this project is the district-level EDA: geography — specifically distance to the CBD and population density — is the strongest structural predictor of both average price levels and cumulative growth rates across districts. District 7 exemplifies how these geographic factors interact to produce exceptional market performance.

References

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